ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C2 · MIXTURES AND CHANGES

Separating and changing

★☆☆
Grade 5

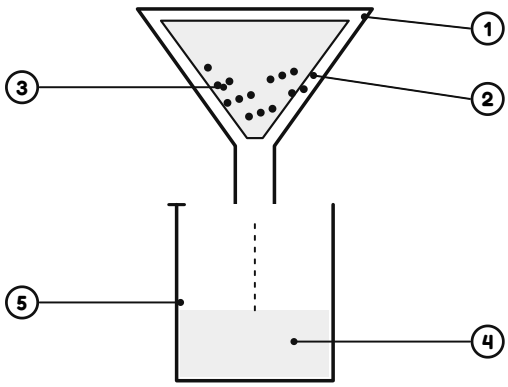
Learning goals

- Choose a separation method.
- Sort reversible and irreversible changes.

1 **Label the filtering equipment.** [5]

LABELS

funnel **filter paper** **clear water (filtrate)** **sand left behind (residue)** **beaker**



1 funnel

2 filter paper

3 sand left behind (residue)

4 clear water (filtrate)

5 beaker

2 **Which method would you use to separate each mixture?** [4]

Mixture	Method
small stones from rice	sieving (or picking out by hand)
iron filings from sand	using a magnet
sand from water	filtering
salt from salty water	evaporating

3 **Sort the materials.** [6]

SORT THESE

salt **sugar** **sand** **flour** **instant coffee** **chalk**

Soluble in water	Insoluble in water
sugar	sand
salt	chalk
instant coffee	flour

4 **Sort the changes.** [6]

SORT THESE

sugar dissolving **ice melting** **frying an egg** **a nail rusting** **water freezing** **burning paper**

Reversible	Irreversible
ice melting	frying an egg
sugar dissolving	burning paper
water freezing	a nail rusting

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C2 · MIXTURES AND CHANGES

Solving separation problems

★★☆
Grade 5

Learning goals

- Plan how to separate a complex mixture.
- Explain dissolving and new materials.

1 **Rana knocked over a jar. Now sand, salt and iron nails are all mixed together. Explain step by step how to get all three back.** [4]

1 Use a magnet to pull out the iron nails. 2 Add water and stir: the salt dissolves, the sand does not. 3 Filter: the sand stays in the filter paper. 4 Evaporate the water from the salty water to get the salt crystals back.

2 **Sumi says, "When sugar dissolves it disappears, so it is gone." Use evidence to explain why she is wrong.** [2]

The water tastes sweet; the solution weighs the same as the water plus the sugar; if you evaporate the water, the sugar is left behind.

3 **How can you tell that baking a cake makes a new material? Give two signs.** [2]

Bubbles of gas make it rise; colour changes to brown; a new smell; you cannot get the eggs and flour back.

4 **Why does filtering not separate salt from salty water?** [2]

The salt is dissolved into tiny pieces that pass through the filter paper with the water.

ANSWER KEY

INVESTIGATE · C2 · MIXTURES AND CHANGES

Does warm water dissolve sugar faster?

★★☆ Grade 5

Learning goals

- Identify variables.
- Plot and read a line graph.
- Make a prediction from a pattern.

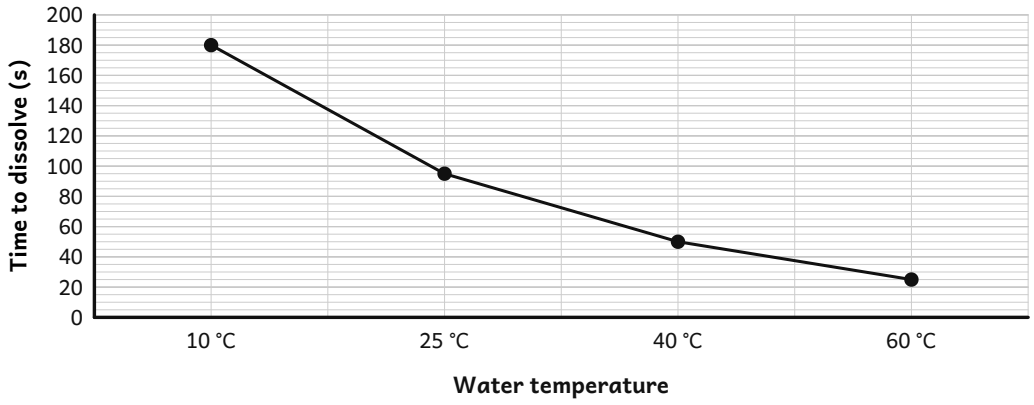
Class 5B timed how long one teaspoon of sugar took to dissolve in 100 ml of water at different temperatures, stirring the same way each time.

1 **Name the variables.** [3]

Change	temperature of the water
Measure	time for the sugar to dissolve (seconds)
Keep the same	amount of sugar and water, type of sugar, stirring, size of cup

2 **Plot the results as a line graph.** [4]

Water temperature (°C)	10	25	40	60
Time to dissolve (s)	180	95	50	25



3 **Describe the pattern.** [2]

The warmer the water, the faster the sugar dissolves (the shorter the time).

4 **Predict the time at 50 °C. Explain how you decided.** [1]

About 35–40 seconds: between the times for 40 °C (50 s) and 60 °C (25 s).

5 **How could they make the results more reliable?** [1]

Repeat each temperature three times and work out an average.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C2 · MIXTURES AND CHANGES

Khejur gur: winter sweetness and a hidden danger

★★★
Grade 5

Learning goals

- Identify reversible and irreversible changes in a real process.
- Explain a health risk using science.
- Communicate a safety message.

Science · Health · Culture · Economics

On cold winter nights in Jashore and other districts, skilled climbers called *gachhi* cut the bark of date palm trees and hang clay pots to collect the sweet **sap**. In the morning the sap is **filtered** through cloth to remove insects and bits of bark, then **boiled** for hours in wide pans. Water **evaporates**, the sap gets thicker, and the sugar turns golden brown. Poured into moulds, it sets as **khejur gur** (date palm molasses), used to make pitha and payesh.

But raw, unboiled sap can be dangerous. Fruit bats sometimes drink from the pots at night and can leave behind the **Nipah virus**, which causes a serious brain illness. Health experts in Bangladesh warn: **never drink raw date palm sap**. Boiling kills the virus, so properly made gur is safe.

Health advice from IEDCR (Institute of Epidemiology, Disease Control and Research), Bangladesh.

1 **Find the separating methods and changes in the story. Complete the table.** [8]

Step	What happens	Reversible or irreversible?
filtering through cloth	removes solid bits (insects, bark) from the sap	reversible (separating)
boiling: water evaporates	water leaves as vapour; the sap gets thicker	reversible (could add water back)
sugar turns golden brown	new substances with new colour and taste form (caramelising)	irreversible
gur sets in moulds	the hot thick liquid cools and becomes solid	reversible (it can be melted again)

2 **Why is filtering through cloth not enough to make raw sap safe to drink?** [2]

Viruses are far too small to be stopped by cloth; only boiling kills them.

3 **Write a short safety message (radio or poster) for families in your area about date palm sap.** [2]

My message

e.g. "Love khejur gur? Enjoy it cooked! Never drink raw date palm sap: it can carry Nipah virus."

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C2 · MIXTURES AND CHANGES

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

J	B	F	W	M	I	U	R	D	T	T
P	U	B	O	A	L	S	N	M	S	S
I	R	K	J	G	S	O	C	I	I	O
W	N	A	F	N	I	L	R	X	E	L
V	A	G	I	E	V	U	P	T	V	U
S	J	U	L	T	H	T	I	U	E	B
O	A	E	T	R	U	I	A	R	R	L
M	O	G	E	G	Y	O	I	E	U	E
H	N	R	R	R	V	N	Y	K	S	M
B	W	S	N	O	E	R	M	O	T	O
O	N	D	I	S	S	O	L	V	E	W

burn **dissolve** **filter** **magnet** **mixture** **rust** **sieve** **soluble** **solution**

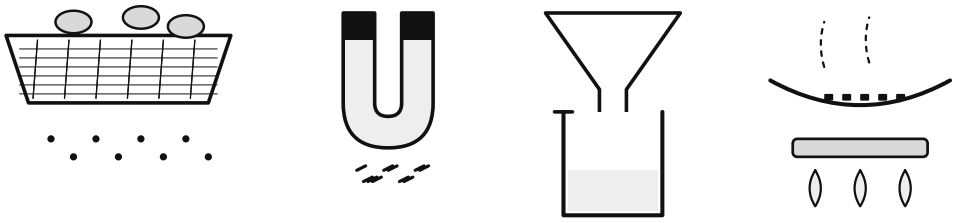
2 **Which of these is an irreversible change?** [1]

- ○ melting chocolate
- ○ dissolving salt
- ● **burning a match**
- ○ freezing juice

3 **Which method separates sand from water?** [1]

- ○ evaporating
- ● **filtering**
- ○ using a magnet
- ○ dissolving

4 **Write the letter of each separation method.** [4]



A B C D

Method	Letter
filtering	C
evaporating	D
sieving	A
magnet	B

5 **Challenge. Is making a cup of cha with sugar and milk a reversible or irreversible change? Explain your thinking.** [2]

Mostly reversible mixing and dissolving (the water could be evaporated), but hot water also draws new flavours out of the tea leaves, so it is hard to fully undo. Accept reasoned answers.